WORKBOOK 05 / 08 · MULTIMEDIA 12

Jump the stream with a designed arc

Anticipation, take-off, breakdowns, landing, overshoot and settle

A JUMP CHANGES POSE AND SUPPORT



1 · ready | 9 · compress | 12 · take-off | 19 · apex | 28 · contact | 36 · recover

Stepped keys first → breakdowns → intentional spacing → recovery.

Jump phase studies: preparation, extension, flight, contact and recovery.

Build: Animate a full jump with readable weight and a controlled landing.

Keep: Jump_36 Graphic + four-to-seven pose strip + revised landing

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

The jump is a change in support: grounded preparation, flight, impact and recovery. The pose and spacing must explain each phase.

Use a provisional 36-frame action: anticipation at 1 and 9; take-off at 12; apex near 19; first landing near 28; settle by 36. These are authored teaching timings, not cues from the old 105-frame lip-sync test. The root travels along an arc while body parts change pose. Uniform easing over the whole jump does not model these phases.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

The stream is wider. Keep the character and goal, then revise anticipation, travel, flight timing and landing reach. Do not stretch the entire picture horizontally.

What you will defend

Show the take-off, apex and first-contact frames and explain one spacing decision.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Plan contact, flight and landing

SELECT: Jump planning strip and a new Jump_36 Graphic.

TOOLS / ROUTE: Guide layer; timeline labels or notes; held pose keys.

1. Mark the near-bank take-off and far-bank landing points on the stream background. Draw a root/hip arc above them.

2. Block standing at 1, deepest anticipation at 9, extended take-off at 12, apex at 19, first landing at 28 and recovered stance at 36.

3. Keep the first version stepped. Check that the landing point is reachable and that the intended take-off direction is clear.

WHY · The action needs a destination and a sequence of support states before it needs smooth movement.

CHECK / SUMMARY · The body clears the stream and lands on the far bank with feet directed toward the ground.

FIX · If the jump appears vertical, increase forward travel and revise the torso lean and leg extension toward the destination.

GLOSSARY · root: A chosen overall body reference, often near the hips.; apex: The highest point of the flight path.

DIFFERENT-OBJECT EXAMPLE · A thrown ball needs a launch point, flight path and destination before its in-betweens are placed.

STRETCH · Design a low athletic jump and a high cautious jump as stepped strips.

02 · Build the anticipation as compression

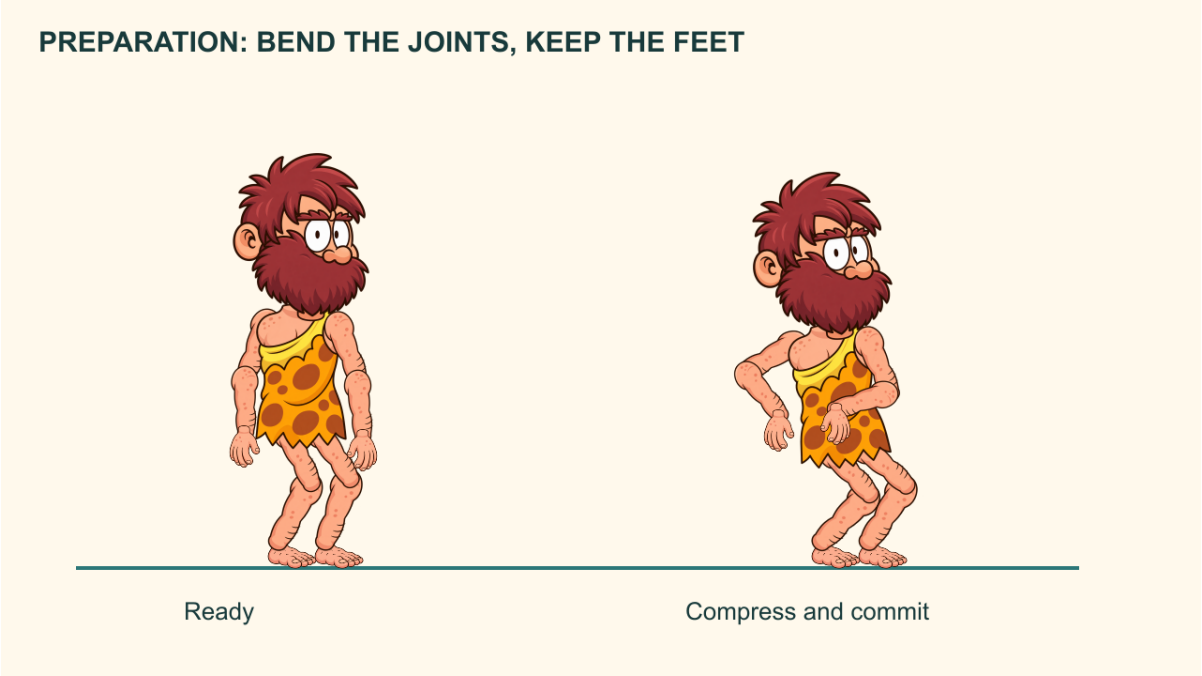
SELECT: Jump_36 frames 1 and 9; all articulated body layers.

TOOLS / ROUTE: Insert Keyframe; Free Transform; ground guide.

1. At frame 9 bend knees and hips, lean the torso toward the destination and move arms back. Keep both feet planted.

2. Lower the hips by posing the legs, not merely scaling the whole Caveman shorter. Keep the head looking where he will land.

3. Use tighter spacing near the deepest crouch if you want a brief commitment before release.



Preparation through bent joints, with feet retained on the ground plane.

WHY · Anticipation stores visual energy and makes the direction of the coming action readable.

CHECK / SUMMARY · The crouch has bent joints, grounded feet and a clear forward intention.

FIX · If the soles lift during the crouch, reposition the hips and knees while preserving foot positions. Reduce the crouch if the rig cannot maintain believable proportions.

GLOSSARY · compression: A pose in which the body folds to prepare or absorb force.; anticipation: The preparation before the main action.

DIFFERENT-OBJECT EXAMPLE · A spring compresses before release; a human jump prepares through joints and weight shift.

STRETCH · Compare a quick dip with a longer uncertain preparation.

03 · Design take-off, an in-flight breakdown and the apex key

SELECT: Jump_36 frames 12, 16 and 19.

TOOLS / ROUTE: Pose keys; Onion Skin; joint repositioning.

1. At take-off extend hips, knees and ankles toward the jump direction. Let the arms swing through to support the action.

2. At frame 16 design an in-flight breakdown between take-off and the apex key: gather the knees and choose the arm path while keeping clear silhouettes. At the frame 19 apex key, show the top-of-flight pose and its different support state.

3. Keep limb lengths consistent. The frame 16 breakdown decides the transition; the apex remains a key. Neither role is defined by being an automatic midpoint.

WHY · The flight pose tells the audience how the character gets across, while take-off explains the launch.

CHECK / SUMMARY · There is a visible change from compressed to extended to gathered, with no disconnected joints.

FIX · If the body looks like a stiff sticker, change the torso angle, knee bend and arm relationship across these phases.

GLOSSARY · extension: Opening the joints into a longer pose.; flight breakdown: A designed in-air pose defining how take-off connects to landing.

DIFFERENT-OBJECT EXAMPLE · A frog extends at launch and gathers its legs during flight rather than keeping one pose throughout.

STRETCH · Let the head lag slightly after take-off, then lead toward the landing.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Space the root along the arc

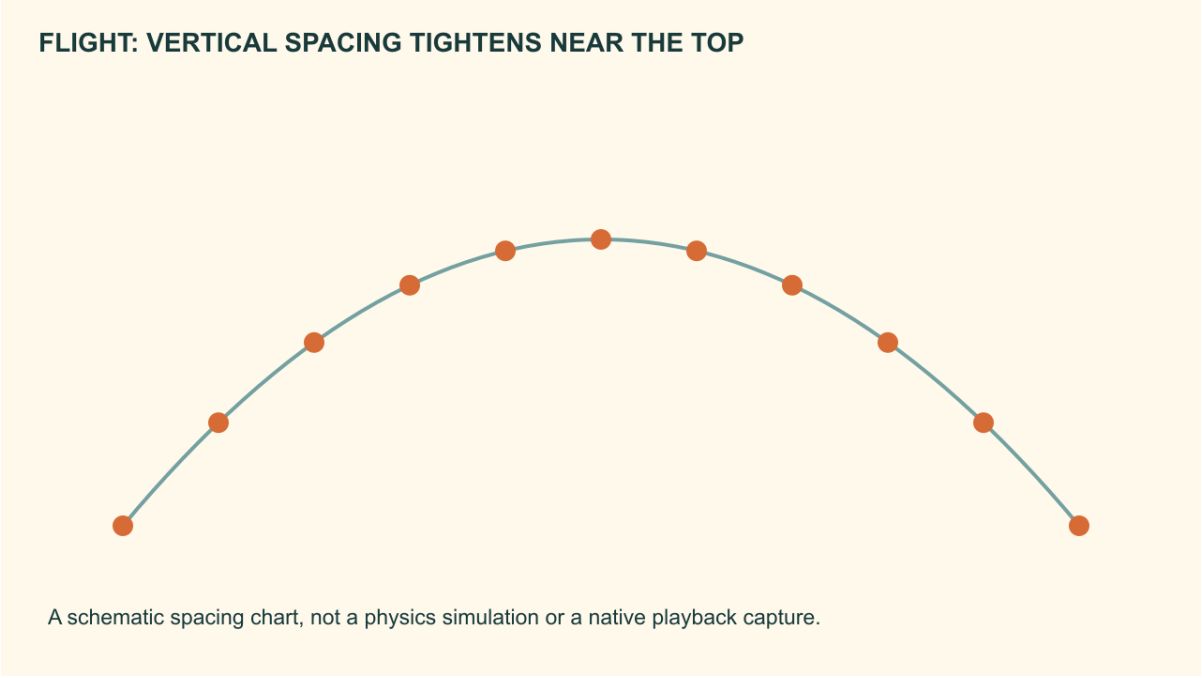
SELECT: Root/whole action placement through frames 12–28.

TOOLS / ROUTE: Inside Jump_36, select all fourteen part-layer keys together for the root offset; keep the outer Stage instance fixed. Guide arc; frame-by-frame preview.

1. Build the root travel inside Jump_36 by moving all fourteen part-layer poses together at each designed frame. Keep its outer Crossing instance fixed during playback. Place intermediate root positions along the planned arc. Use smaller vertical changes near the apex and larger vertical changes during faster ascent/descent.

2. Keep forward travel reasonably consistent through flight unless your deliberate style calls for otherwise. Do not apply a single strong ease-in/ease-out to the entire jump.

3. At each important root position, repose limbs as required. Inspect the trajectory in context with the stream banks.



Schematic trajectory study: vertical displacement per frame becomes smaller near the apex. Not a physics simulation.

WHY · Timing controls flight duration; spacing communicates changing speed. The body also changes shape while following the path.

CHECK / SUMMARY · The jump rises and falls smoothly, slows vertically near the top and does not hover accidentally.

FIX · If the path makes a sharp corner, move the intermediate root point onto a continuous curve. If the landing floats, diagnose a pause or slow descent: reduce an unintended hold, increase downward position spacing toward impact where appropriate, and correct the actual contact pose.

GLOSSARY · trajectory: The path of an object or body through space.; spacing chart: A plan showing successive positions along a path.

DIFFERENT-OBJECT EXAMPLE · A ball moves only a little vertically near its apex but much farther per frame lower in its arc.

STRETCH · Keep the same arc but change flight duration and describe the change in apparent weight.

05 · Absorb the landing, then settle

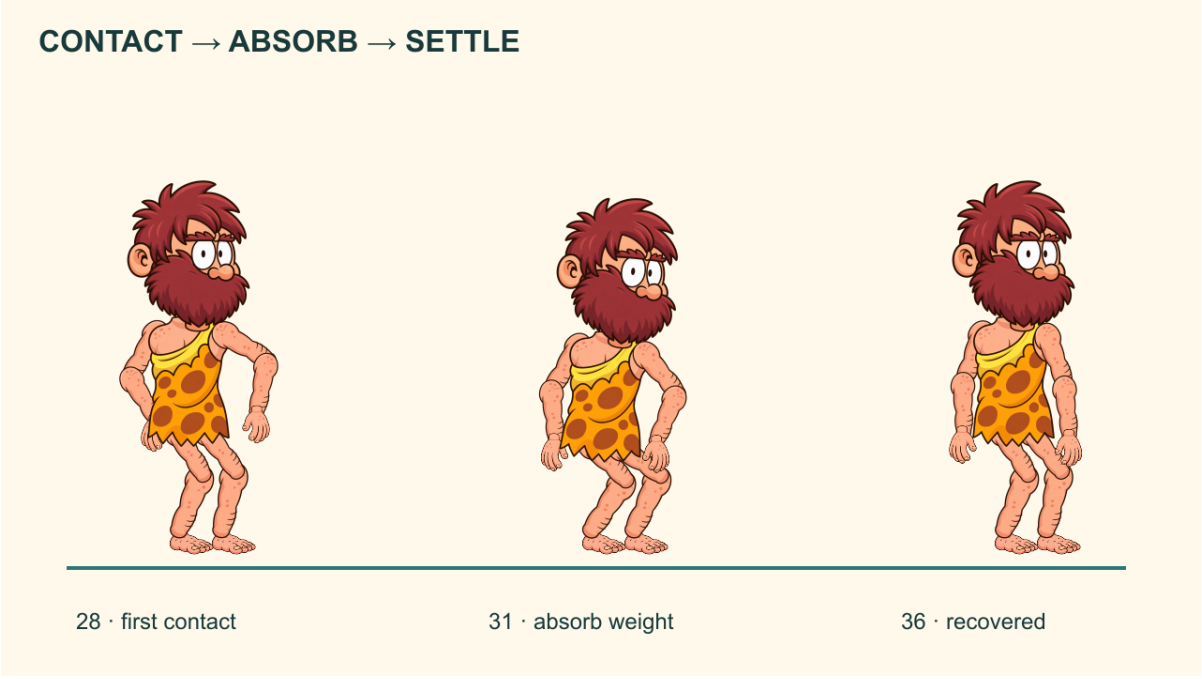
SELECT: Frames 28, 31 and 36; feet and hips visible.

TOOLS / ROUTE: Insert Keyframe; Free Transform; contact guide.

1. At first contact, place the leading foot on the far bank. At 31 bend knees and hips to absorb the landing; keep the contact foot fixed.

2. Let the torso and arms continue slightly past their final pose, then recover by 36 with a smaller movement.

3. Use restrained squash/stretch only as an optional accent. Preserve apparent body volume and facial readability; joint posing must still explain the landing.



First contact, absorption and decreasing recovery are separate pose jobs.

WHY · Impact requires a response. A straight-legged frozen landing removes weight and can look painful.

CHECK / SUMMARY · The contact foot stays grounded, the body compresses after contact and the recovery decreases in size.

FIX · If the Caveman bounces repeatedly, reduce the overshoot and number of recovery keys. If he sinks through the bank, restore contact before adjusting the torso.

GLOSSARY · squash and stretch: Controlled shape change that suggests force and flexibility while preserving apparent volume.; overshoot and settle: A small pass beyond rest followed by decreasing recovery.

DIFFERENT-OBJECT EXAMPLE · A soft ball compresses at impact; a heavy box may show its impact through a brief recoil and a strong stop instead.

STRETCH · Make a heavy landing without enlarging the character's head or changing its identity.

06 · Connect the jump to retrieving the stone

SELECT: A named 36-frame Pickup Graphic after the settled landing; stone contact separate from body parts.

TOOLS / ROUTE: New pickup pose keys; hand and prop contact.

1. Create a 36-frame Graphic named Pickup from the character template. Match local frame 1 to the settled jump pose. Bend through knees/hips and reach the hand to the stone; use local frame 25 as the worked grasp cue, then lift and recover by 36.

2. Keep the bank stone on the main Props layer until the grasp. At Pickup local 25 (Crossing 145 in the worked map), blank the bank copy and reveal a stone inside Pickup aligned to the hand. From then on animate that carried copy with the hand. Avoid displaying both copies.

3. Finish with a clear held-stone pose that supports the later cut back to the cave.

WHY · The jump serves the story goal. Retrieval proves the action achieved something.

CHECK / SUMMARY · The stone remains on the bank until contact and stays held afterward.

FIX · If the stone moves before the hand reaches it, move the prop key to the actual grasp frame. If it slides in the hand, inspect their relative positions.

GLOSSARY · contact frame: The precise frame at which two elements first meet.; continuity of props: Keeping an object's location and ownership understandable.

DIFFERENT-OBJECT EXAMPLE · A cup remains on a table until the fingers contact it, then follows the lift.

STRETCH · Add a brief effort pose before lifting a heavier-looking stone.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair a weightless jump

SELECT: A duplicate using one unchanged body pose along a straight line.

TOOLS / ROUTE: Compare playback; restore designed keys and arc.

1. Play the deliberately poor version. Identify two causes: an unchanged pose, missing anticipation, uniform spacing, no arc or an unabsorbed landing.

2. Repair those two causes without adding decorative effects. Keep a before/after pose strip.

3. Explain which change improves intention and which improves weight.

WHY · Principles are tools for diagnosing an action, not labels to add after it is finished.

CHECK / SUMMARY · The repair shows phase changes and a grounded landing; the explanation cites visible frames.

FIX · If the result is only smoother, return to the held keys and strengthen the support changes.

GLOSSARY · weight: The impression of mass created by timing, pose, support and response to force.

DIFFERENT-OBJECT EXAMPLE · A bowling ball and a balloon can follow similar routes but require very different spacing and recovery.

STRETCH · Reduce the number of keys while keeping all four support phases readable.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the wider-stream version

SELECT: Your independent Jump_Wide Graphic and matching background staging.

TOOLS / ROUTE: Duplicate action; revise keys, travel and timing.

1. Widen the gap by an amount your character can plausibly clear. Revise the preparation, flight route and landing pose together.

2. Keep a labelled anticipation key, take-off key, flight breakdown, first-contact key and settle. Add intentional in-betweens.

3. Save and reopen. Show the two versions and defend one timing or spacing decision. Choose the version that belongs in the film.

WHY · A new distance requires a new movement solution and an explanation of the trade-off.

CHECK / SUMMARY · The character clears the changed obstacle and retains believable support and proportion.

FIX · If you need impossible reach, revise the staging or introduce a deliberate stylized exaggeration and justify it; do not silently stretch anatomy.

GLOSSARY · exaggeration: A purposeful amplification that makes an idea clearer while preserving its logic.

DIFFERENT-OBJECT EXAMPLE · Independent task: use the same support and arc criteria; no wider-gap pose solution is supplied.

STRETCH · Make the wider jump feel risky without adding extra dialogue.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

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